

ME 333
MECHANICS OF DEFORMABLE BODIES

Lecture 21: Shear Center II &
Tutorial Problems

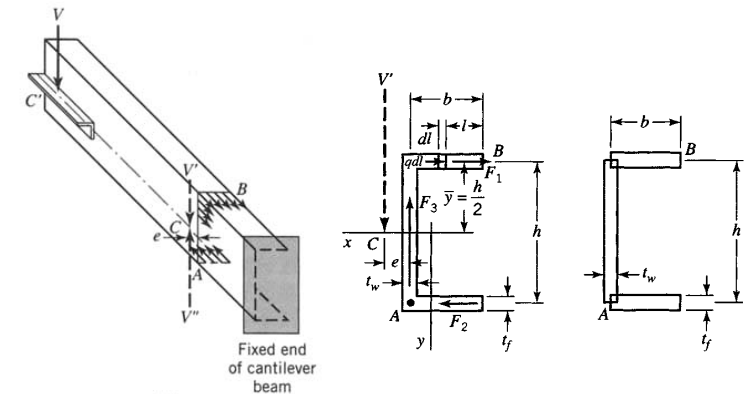
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Sem I, 2025 - 2026

Shear Center

Shear Center for a channel section:



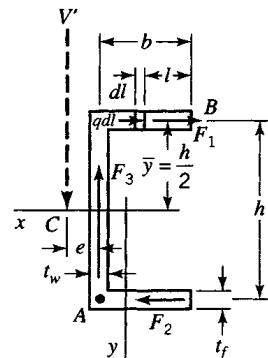
Ref Base Fig: Advanced Mechanics of Materials, Boreisi & Schmidt

Shear Center

Shear Center

Shear Center for a channel section

Objective : Determine the distance e and locate the shear center C .



- Static force and moment equilibrium:

$$\begin{aligned}\sum F_x = F_2 - F_1 &= 0 \implies F_1 = F_2 \\ \sum F_y = V' - F_3 &= 0 \implies F_3 = V' \\ \sum M_A = V'e - F_1h &= 0 \implies e = \frac{F_1h}{V'}\end{aligned}$$

Q How to determine F_1 ?

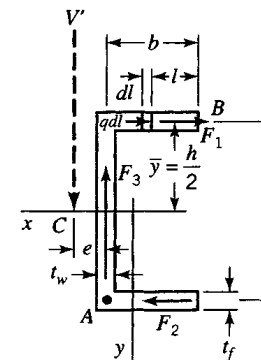
Ref Base Fig: Advanced Mechanics of Materials, Boreisi & Schmidt

Shear Center

Shear Center

Shear Center for a channel section

- Determine the force F_1 :



$$\begin{aligned}F_1 &= \int_0^b q \, dl = \frac{V'}{I_{xx}} \int_0^b A \bar{y} \, dl \\ &= \frac{V' t_f h}{2 I_{xx}} \int_0^b l \, dl \\ &= \frac{V' t_f b^2 h}{4 I_{xx}}\end{aligned}$$

Q What is I_{xx} ?

Ref Base Fig: Advanced Mechanics of Materials, Boreisi & Schmidt

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Shear Center for a channel section

- Calculate Moment of Inertia (I_{xx}) :

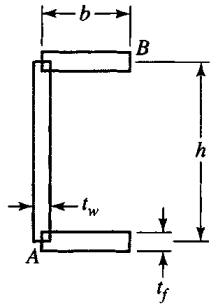
$$I_{xx} = \frac{1}{12} t_w h^3 + 2 b t_f \left(\frac{h}{2} \right)^2 + 2 \frac{1}{12} b t_f^3$$

$$\approx \frac{1}{12} t_w h^3 + 2 b t_f \left(\frac{h}{2} \right)^2 \quad (\because t_f \ll h)$$

- Calculate the distance e :

$$e = \frac{F_1 h}{V'} = \frac{V' t_f b^2 h^2}{4 I_{xx} V'} = \frac{t_f b^2 h^2}{4 \left(\frac{1}{12} t_w h^3 + 2 b t_f \left(\frac{h}{2} \right)^2 \right)}$$

$$= \frac{b}{2 + \frac{t_w h}{3 t_f b}}$$



Ref Base Fig: Advanced Mechanics of Materials, Borei & Schmidt

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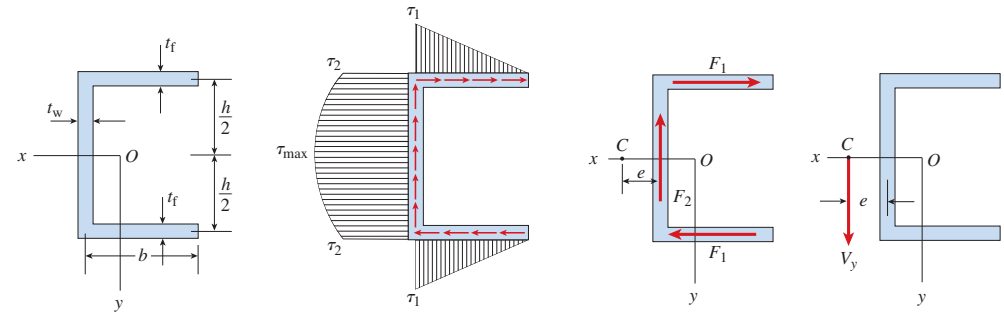
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Shear Center for a channel section:



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Shear Center

Summary of Definitions :

- Any plane on which the externally applied load acts is called the *Plane of Loading*.
- The line joining centroids of all cross sections along the axis of the beam is called as *Centroidal Axis*.
- If the orientation of the *Centroidal Axes* is such that the product moment of area of the cross-section is zero it is called *Principal Centroidal Axes*.
- Shear Center* is the point about which if an external load is applied it produces no twisting moment.
- A line joining the shear centers of all cross sections along the axis of the beam is called *Bending Axis*.

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Shear Center

Key Points :

- Moments are resolved about the *Centroidal axes*
- Shear forces are resolved about the *Bending axes*
- Shear center lies always on the line and in the plane of symmetry.
- For a beam with cross-section that has two or more axes of symmetry the bending axis is same as the longitudinal centroidal axis.
- For cross-sections with one axis of symmetry, the shear center and centroid do not coincide.
- Bending axis is parallel to the centroidal axis of the beam. They do not coincide except in very special cases.

Ref Base Fig: Advanced Mechanics of Materials, Borei & Schmidt

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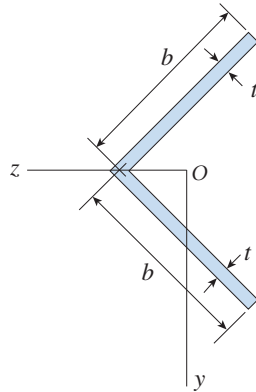
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[Example 1]: Determine the shear center.



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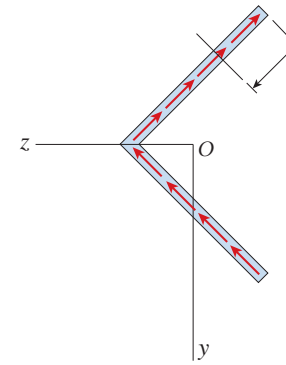
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[Problem 1] Contd ...



- Step 1 : Static force and moment equilibrium
- Step 2 : Determine the force F_1
- Step 3 : Calculate I_{xx} and distance e

Q Do we need the 3 step analysis ?

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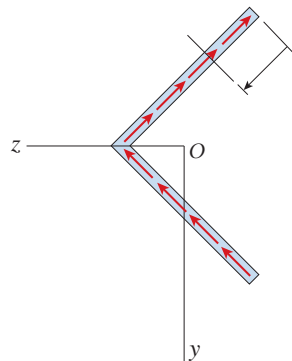
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[Example 1] Step 1 : Static Equilibrium



- Static force and moment equilibrium:

$$\sum F_x = F_2 - F_1 = 0 \implies F_1 = F_2$$

$$\sum F_y = V' - F_3 = 0 \implies F_3 = V'$$

$$\sum M_A = V'e - F_1 h = 0 \implies e = \frac{F_1 h}{V'}$$

Q How to determine F_1 ?

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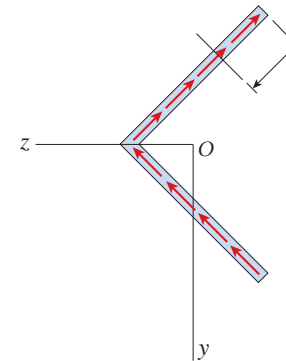
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[Problem 1] Contd ...



- Step 2 : Determine the force F_1 :

$$F_1 = \int_0^b q dl = \frac{V'}{I_{xx}} \int_0^b A \bar{y} dl = \frac{V' t_f h}{2 I_{xx}} \int_0^b l dl = \frac{V' t_f b^2 h}{4 I_{xx}}$$

- Step 3 : Calculate I_{xx} and distance e :

$$I_{xx} = \frac{1}{12} t_w h^3 + 2 b t_f \left(\frac{h}{2} \right)^2 ; \quad e = \frac{b}{2 + \frac{t_w h}{3 t_f b}}$$

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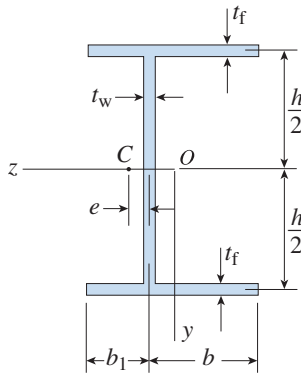
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[Problem 2]: Determine the shear center.



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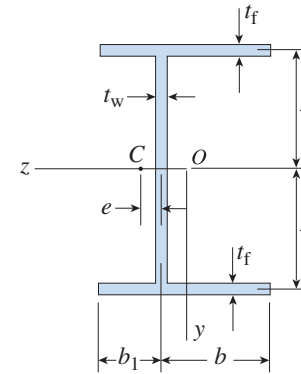
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[Problem 2] Contd ...



- Step 1 : Static force and moment equilibrium

$$\sum F_z = -F_1 + F_2 + F_5 - F_4 = 0 \implies F_1 = F_5; F_2 = F_4$$

$$\sum F_y = V' - F_3 = 0 \implies F_3 = V'$$

$$\sum M_A = V'e + F_5h - F_4h = 0 \implies e = \frac{(F_5 - F_4)h}{V'}$$

- Note: F_1 (bottom) and F_5 (top) are acting in the flange section b_1 creating a counter clockwise flow, while F_2 and F_4 are in the flange section b producing a clockwise flow. V is acting downwards at C . Point A is at the junction connecting the lower flange to the web.

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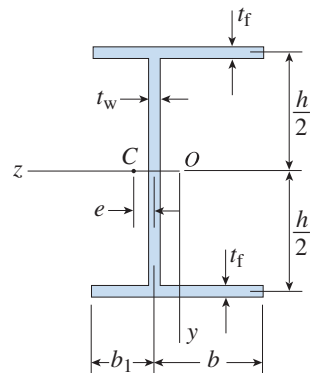
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Ref Base Fig: Mechanics of Materials, Gere & Goodno

Shear Center

[Problem 2] Contd ...



- Step 2 : Determine the unknown forces

$$F_4 = \int_0^b q dl = \frac{V'}{I_{zz}} \int_0^b A \bar{y} dl = \frac{V' t_f h}{2 I_{zz}} \int_0^b l dl = \frac{V' t_f b^2 h}{4 I_{zz}}$$

$$F_5 = \int_0^{b_1} q dl = \frac{V' t_f b_1^2 h}{4 I_{zz}}$$

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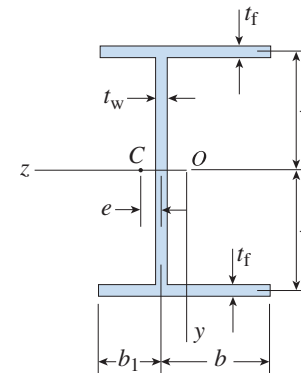
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[Problem 2] Contd ...



- Step 3 : Calculate I_{zz} and distance e

$$I_{zz} = \frac{1}{12} t_w h^3 + 2(b + b_1) t_f \left(\frac{h}{2} \right)^2$$

$$V'e = (F_4 - F_5)h = \frac{V' t_f (b^2 - b_1^2) h}{4 I_{zz}}$$

$$\implies e = \frac{b \left(1 - \frac{b_1^2}{b^2} \right)}{\left[2 + 2 \frac{b_1}{b} + \frac{t_w h}{3 t_f b} \right]}$$

Ref Base Fig: Mechanics of Materials, Gere & Goodno

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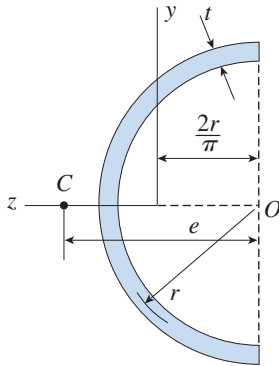
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[Problem 3]: Determine the shear center.



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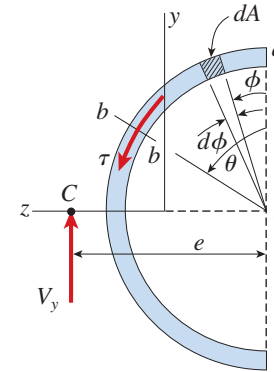
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[Problem 3] Contd ...



- Step 1 : Static force and moment equilibrium

$$\sum M_O = -V_y e + (\tau dA) r = 0$$

$$\Rightarrow e = \frac{(\tau dA) r}{V_y}$$

- Step 2 : Determine Q & I_{zz}

$$Q = \int y dA = \int (r \cos \phi) r t d\phi$$

$$I_{zz} = \frac{\pi r^3 t}{2}$$

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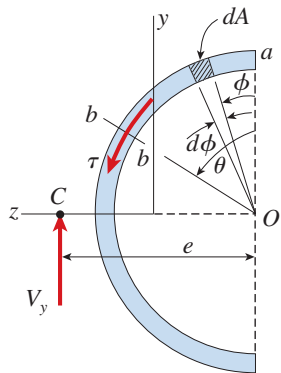
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[Problem 3] Contd ...



- Step 3 : Determine τ

$$\tau = \frac{q_{xs}}{t} = \frac{V_y Q}{I_{zz} t}$$

- Step 4 : Determine e

$$e = \int_0^\pi \frac{(\tau dA) r}{V_y} = \frac{4r}{\pi}$$

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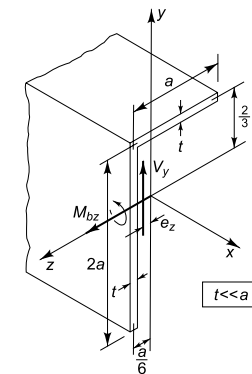
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Shear Flow

[Problem 4]: [Example 7.10] Determine the distribution of shear flow. Given $I_{zz} = \frac{4}{3}a^3t$,
 $I_{yy} = \frac{1}{4}a^3t$, $I_{yz} = -\frac{1}{3}a^3t$.



Ref Base Fig: An Introduction to the Mechanics of Solids, Crandall et al.

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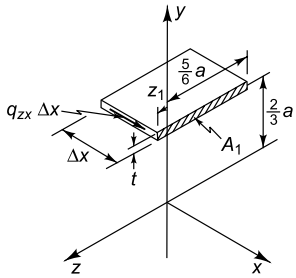
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Shear Flow

[Problem 4] Contd ...

$$q_{xz} = q_{xs} = \frac{-V_y}{I_{zz}I_{yy} - I_{yz}^2} \left(I_{yy} \int_{A_1} y dA - I_{yz} \int_{A_1} z dA \right)$$



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Reading Assignment

Chapter 7 : Crandall et. al.
Chapters 7 & 8 : Boresi & Schmidt